Curriculum Vitae: Samuel John Dunham

samueljdunham@protonmail.com

Research Experience

California Institute of Technology, Pasadena CA
Postdoctoral Scholar, 06/2024 - 05/2026

- Developing numerical method for solving visco-resistive GRMHD equations inspired by 14-moment methods on GPUs for execution on HPC clusters

Vanderbilt University, Nashville TN
Research Assistant, 08/2016 - 05/2024

- Developed solver for compressible general relativistic hydrodynamics equations with Runge–Kutta discontinuous Galerkin methods for execution on leadership-class supercomputers
- Ported and optimized code developed for CPUs to run on GPUs via OpenACC and OpenMP Offloading
- Coupled code to AMReX
- Developed AMReX code to allow its adaptive mesh refinement tools to work with DG methods

University of Michigan, Ann Arbor MI
Research Assistant, 06/2014 - 05/2016

- Analyzed data for multiple images of background sources due to strong gravitational lensing by galaxy clusters
- Found robust lens models for several galaxy clusters, from which was deduced the mass of the cluster core, the total magnification provided by the cluster, the location of the source, and its morphology

Education

Vanderbilt University, Nashville TN
Ph.D. Astrophysics, May 2024
Thesis title: THORNADO-HYDRO, XCFC: A DISCONTINUOUS GALERKIN HYDRODYNAMICS SOLVER FOR GENERAL RELATIVISTIC CORE-COLLAPSE SUPERNOVA SIMULATIONS

Fisk University, Nashville TN
M.A. Physics, December 2018

University of Michigan, Ann Arbor MI
B.S. Astronomy and Astrophysics, May 2016
B.S. Interdisciplinary Physics with Astronomy, May 2016

- Graduated Magna Cum Laude

Washtenaw Community College, Ann Arbor MI
Associate's Degree in General Studies in Math and Natural Science, May 2013

Fellowships/Grants

- Earned McMinn summer research fellowship for outstanding students in the Department of Physics and Astronomy (May 2020)

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- Earned honors grant for poster presentation at American Astronomical Society (AAS) conference (January 2016)

Training/Development

Attended Binary Neutron Star Merger Summer School, 05/16/2018 – 05/18/2018, Michigan State University, East Lansing, MI

Attended Statistics Workshop for Astronomers, 05/05/2017 – 05/11/2017, Vanderbilt University, Nashville, TN

Publications

Samuel J. Dunham & Elias Most, "General-Relativistic Visco-Resistive-Diffusive-Hall Magnetohydrodynamics: Numerical Method and Tests" (in prep.)

Samuel J. Dunham et al., "thornado-HydroXCFC: A Discontinuous Galerkin Hydrodynamics Solver for Core-Collapse Supernova Simulations for the Extended Conformally-Flat Condition" (in prep.)

Samuel J. Dunham et al., "A Parametric Study of the SASI Comparing General Relativistic and Nonrelativistic Treatments", (2024) ApJ, **964**:38

David Pochik et al., "thornado-hydro: A Discontinuous Galerkin Method for Supernova Hydrodynamics with Nuclear Equations of State", (2021) ApJ, **253**:21

Samuel J. Dunham et al., "A discontinuous Galerkin method for general relativistic hydrodynamics in thornado", (2020) J. Phys.: Conf. Ser. **1623**:012012

Keren Sharon et al., "Strong Lens Models for 37 Clusters of Galaxies from the SDSS Giant Arcs Survey", (2020) ApJ, **247**:12

Eirik Endeve et al., "thornado-hydro: towards discontinuous Galerkin methods for supernova hydrodynamics", (2019) J. Phys.: Conf. Ser. **1225**:012014

Samuel J. Dunham et al., "Lens Model and Source Reconstruction Reveal the Morphology and Star Formation Distribution in the Cool Spiral LIRG SDSS J1438+1454", (2019) ApJ, **875**:18

Presentations

"thornado-Hydro (xCFC)", AAS 243, January 2024, oral (dissertation talk)

"A Parametric Study of the SASI Comparing General Relativistic and Non-Relativistic Treatments", MICRA 2023, oral

"A Discontinuous Galerkin Method for General Relativistic Hydrodynamics in `thornado`", AstroNum 2019, oral

"A Discontinuous Galerkin Method for General Relativistic Hydrodynamics in `thornado`", APS April 2019, oral

“A Discontinuous Galerkin Method for General Relativistic Hydrodynamics”, APS April 2018, poster

“Strong Lens Models for 10 Galaxy Clusters from the Sloan Giant Arcs Survey”, AAS January 2016, poster

Outreach

- Co-organized a week-long Caltech Relativistic Astrophysics Summer School (June 2025)
- Led a two-week computational bootcamp for incoming Fisk-Vanderbilt Masters-to-PhD Bridge students (August 2022)
- Participated twice in Skype-a-Scientist (2020)

Skills

- Expertise in theory and computational modeling of viscous, resistive, relativistic plasmas
- Expertise in 14-moment methods for dissipative (magneto-)hydrodynamics
- Expertise with coding in Python, C++, and Fortran90
- Experience with coding in Julia
- Experience working with and developing code written by others
- Experience solving PDEs on computers
- Proficiency submitting and monitoring jobs on leadership-class supercomputers
- Expertise with programming/optimizing code for GPUs via OpenACC and OpenMP offloading
- Experience working with distributed memory systems on heterogeneous architectures
- Expertise with Linux, especially Arch Linux

Misc

- GitHub page: `https://www.github.com/dunhamsj`
- Personal website: `https://www.samueljdunham.com/`